MIL-M-27579E (USAF)
09 July 1985
SUPERSEDING
MIL-T-27579D (USAF)
17 December 1970

MILITARY SPECIFICATION

MANUALS, TECHNICAL: AIRCREW, NUCLEAR WEAPON (BOMB) DELIVERY, MULTIMODE, PREPARATION OF

1. SCOPE

1.1 This specification provides detailed instructions for the preparation of technical manuals and checklists for multimode delivery of nuclear weapons (bombs) from tactical aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a port of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-38778	Checklists: Title Page, List of Effective Pages, Printing, and Binders, General Requirements for Preparation of
MIL-M-38784	Manuals, Technical: General Requirements for Preparation of
MIL-P-38790	Manuals, General Requirements for (Photolithographic Negatives, Reproduction Assembly Sheets, Printing)

TECHNICAL ORDERS

00-5-1	Air Force Technical Order System
00-5-2	Technical Order Distribution and Requisition System

TMSS

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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3. REQUIREMENTS

3.1 General Requirements

3.1.1 General Specifications. The requirements of MIL-M-38784, MIL-C-38778, and MIL-P-38790 apply as requirements of this specification to the extent specified herein. Unless otherwise specified in the contract, the technical manual(s) and checklist(s) shall be submitted in the form of negatives.

3.1.1.1 Lists. List of tables, charts, illustrations and Time Compliance Technical Orders (TCTO) shall not be required.

3.1.1.2 Illustrations. Illustrations may be used more than once, providing such use is essential. Figure numbers and titles shall not be used in the normal aircrew procedures section(s). If an illustration is required to emphasize a specific step, the step number shall be placed below the illustration. In other than the procedures, figure titles may be placed with the illustrations. In these cases, the figure number (and classification, if applicable) shall be placed below the figure.

3.1.1.3 Revisions. Change bars shall not be required in revisions.

3.1.1.4 Bomb Tables. Bomb tables (charts) shall not carry table numbers. Reference to bomb tables shall be to the applicable page number only. Bomb tables may be tabbed by black, bleed tabs; approximately ½ inch high. The tables shall be preceded by an index page that identifies the tabs or a table of contents page. Bomb table pages shall be identified by pertinent parameters such as bomb designations, and delivery mode; this identification shall be placed below the TO number.

3.1.1.5 Nomenclature. Nomenclature shall be standardized throughout the manual. If authorized, nomenclature may be shortened or abbreviated to its common name.

3.1.1.6 Flagging. Flagging may be used in the Technical Orders (TO) for manuals and checklists to denote items applicable to specific aircraft models, daily checks, etc. Any flagging used shall be explained in the TO introduction. Flagging normally will be a parenthesized entry; e.g., (P), (RO), etc.

3.1.2 Text Presentation. Text shall be presented with unjustified right hand margins. Standard office typewriters may be used to prepare reproducible copy; final type size for the text shall be approximately 10 point (pt.). Headings for Note, Caution, and Warning shall follow the first examples in MIL-M-38784.

3.1.3 Classified Manuals. Classified manuals for US use shall be marked as specified in MIL-M-38784. Classified manuals intended for non-US use shall be marked as specified in the contract.

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3.1.4 Type of Manual and Checklist to be Prepared. The aircrew bomb delivery technical manual shall normally be multivolume type, consisting of the following elements:

Volume I	- Basic Information	
Volume II	- Aircrew Procedures and Supplementary Information	8 1/4 by 10 3/4 inches, trim size
Volume III	- Mission Planning	
Checklist(s), 4 1/2 by 7 inches, trim size		

A single volume manual (8 1/4 by 10 3/4 inches, trim size), with checklist(s) (4 1/2 by 7 inches, trim size) will be prepared for practice bomb delivery manuals (3.4 herein) and manuals for non-US use (3.3 herein).

3.2 Arrangement (Multivolume Manual). Volumes I, II, and III shall have the following arrangement:

Front Matter

Title Page

List of Effective Pages

Table of Contents

Safety Precautions (Volume II only)

Introduction

Section I, II, etc., - Instructions

Alphabetical Index

NOTE: Section titles within each volume are listed in specific requirements for the volume. If no data are available for a specific section, that section may be omitted and succeeding sections renumbered.

3.2.1 Front Matter. The front matter shall be prepared as described in the following paragraphs.

3.2.1.1 Title Page. Title pages and checklists shall be prepared in accordance with the requirements of MIL-M-38784. Title pages shall contain no classified information. Specific titles and aircraft/bomb designations shall normally be as follows:

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Aircrew Bomb Delivery Basic Information (or Procedures or Mission Planning, as applicable)

Bxx/Bxx (as applicable) Bombs

USAF Series

Aircraft Designation

Contract Numbers

Title and aircraft/bomb information shall be centered horizontally below the words TECHNICAL MANUAL. For checklists, the word CHECKLIST shall replace TECHNICAL MANUAL. Since the equipment covered is applicable to more than one contractor, contractor's identification shall not be used.

3.2.1.1.1 Transmittal Notice. The following distribution limitation statements shall be placed on each title page:

THIS PUBLICATION IS REQUIRED FOR OFFICIAL USE OR FOR ADMINISTRATIVE OR OPERATIONAL PURPOSES ONLY. DISTRIBUTION IS LIMITED TO U.S. GOVERNMENT AGENCIES. OTHER REQUESTS FOR THIS DOCUMENT MUST BE REFERRED TO SAN ANTONIO AIR LOGISTICS CENTER/SWPPT, KELLY AFB, TEXAS 78241.

3.2.1.2 List of Effective Pages. The list of effective pages shall be prepared in accordance with MIL-M-38784.

3.2.1.3 Table of Contents. Each manual (or volume thereof) shall have a table of contents listing the section titles and their applicable page numbers. In addition, the first page of each section shall include a table of contents listing the major headings within the section. Text shall begin immediately below the section table of contents. Tables of contents shall normally be typed single column.

3.2.1.4 Safety Precautions. Volume II shall contain safety precautions as the first item in Section I. Safety precautions will then be listed as the first item in the table of contents. The data shall include safety precautions applicable to the bombs and the weapon system, and applicable definitions. (NOTE: The safety precautions entry is not required for practice bomb manuals or manuals for non-US use.)

3.2.1.5 Introduction. The introduction shall be prepared in accordance with the requirements of MIL-M-38784. Also the following additional requirements apply:

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a. List of applicable reference documents. This list shall be held to a minimum, showing essential documents only.

b. Statements concerning technical responsibility for the manual and a submission of change requests and comments.

3.2.2 Content (Volume I). Volume I (Basic Information) shall contain the following data.

Section I - Description

Section II - Bomb Preflight Procedures

Volume I will be common to all MDS in the weapon system series.

3.2.2.1 Section I - Description. This section shall contain, but is not limited to, the following data:

a. Nuclear bomb deliveries

b. Nuclear effects data

c. Nuclear suspension equipment

d. Other aircraft equipment common to all aircraft mods

The weapon descriptive data requirements listed above shall be aircrew oriented. Appropriate notes, cautions, and warnings as required to assure safety of operations shall be included.

3.2.2.2 Section II - Bomb Preflight Procedures. This section shall contain bomb preflight procedure for strike or ferry missions. The procedures shall be illustrated. Format will be as detailed in para 3.2.3.3.

3.2.2.3 Alphabetical Index. Volume I shall contain an alphabetical index or table of contents. This index shall be prepared in accordance with the general requirements of MIL-M-38784; it shall list pertinent items and their page numbers. If more than one page in length, the alphabetical index shall be prepared in double column.

3.2.3 Content (Volume II). Volume II (Procedures) may contain data applicable to only one MDS in the aircraft series. Data applicable to more than one MDS may be included in one manual if required by the procuring activity.

Section I - Safety Precautions, Radio Calls, and Unscheduled Landing Procedures.

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Section II - Bomb Monitor, Control and Release Systems (descriptive material).

Section III - Normal Aircrew Procedures.

Section IV - Emergency Aircrew Procedures.

Only procedures that specifically apply to nuclear bombs shall be included.

3.2.3.1 Section I, Safety Precautions, Radio Calls, and Unscheduled Landing Procedures. This section may contain other required information as directed by the procuring activity.

3.2.3.2 Section II, Bomb Monitor, Control and Release. This section shall contain descriptions of the systems. Cockpit switch functions may be explained in tables test.

3.2.3.3 Section I, ( ), Normal Aircrew Procedures. The normal aircrew procedures, presented in the form of illustrated checklists, shall contain the procedures to be followed from preflight (except bomb preflight) through postmission for each control-monitor, with all applicable delivery modes and bombs covered within the section. However, if the procedural differences between control-monitors are few, the procedures may be integrated and placed in a single section. The text for each procedural step consists of an unclassified demand-response line for the step, followed by an amplification, if required. Amplification may contain classified material. Within the normal aircrew procedures, the portion for initiation and execution of bombing run may be in narrative form instead of step-by-step procedures, presented separately for each mode of delivery (such as LADD, Level). Blocks of procedures (within headings and subheadings) shall normally be separated by a horizontal line (3 to 6 pt) across the page. Procedural steps shall be numbered in accordance with Illustations, if used, shall be as small as practical and shall preferably be placed along the right side of the page, and shall be keyed to the applicable step number(s). Pages shall be identified by pertinent parameters such as bomb designations and delivery mode; this identification shall be placed below the TO number. The following breakdown shall be used within each section:

Part I - Strike Procedures

Part II - Ferry and Deployment Procedures, as applicable.

3.2.3.3.1 Part I - Strike Procedures. Part I shall contain the following items:

Preflight Procedures (except bomb preflight)

Ground Alert

Launch Procedures

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Inflight Procedures

(Mode)

(Mode)

Abort Procedures

3.2.3.3.1.1 Preflight Procedures. This portion shall include all procedures that must be accomplished in the cockpit prior to ground alert or launch.

3.2.3.3.1.2 Ground Alert Procedures. This portion shall contain information and/or procedures relative to placing an aircraft on alert, alert crew changeover and instructions in aircraft operations while on alert.

3.2.3.3.1.3 Launch Procedures. This portion shall include for each control-monitor all procedures that must be accomplished prior to takeoff.

3.2.3.3.1.4 Inflight Procedures. This portion shall include for each delivery mode instructions for all inflight procedures; these instructions shall be presented in the following order:

After takeoff

Bomb Line (or predetermined point)

Prior to bombing run

Initiation and execution of bombing run (for each delivery mode)

After escape maneuver

3.2.3.3.1.5 Abort Procedures. This portion shall include procedures to be followed in the case of an aborted mission of an aircraft carrying a nuclear bomb which has been prepared for release. These procedures shall be presented in the following order:

At time of abort

Prior to landing

After landing (if required)

3.2.3.3.1.6 Part II - Ferry and Deployment Procedures. Part II will contain Preflight Procedures only for deployment and for tactical ferry, as applicable. Radio calls and transfer procedures will be handled by notes only and not be detailed information.

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3.2.3.4 Section IV - Emergency Procedures. Emergency aircrew procedures shall be presented in the form of an amplified checklist. Each checklist step shall consist of a demand-response line, followed by an amplification if required. Emergency situations shall be outlined and possible cause and corrective action shall be given. The following topics shall be covered.

Emergency release procedures

Bailout procedures with bomb aboard

Aircraft electrical system failure

Bomb malfunction analysis

Bomb involved in fire or ground accident

Ground operations and barrier engagement

Crashlanding

Bomb malfunction analysis shall be presented in tabular form; these tables are normally presented following the complete text of the section.

3.2.3.4.1 Marking. Emergency pages shall be marked as specified in MIL-M-38784.

3.2.4 Checklists. Checklists, based on the aircrew procedures (see 3.2.2.2 and 3.2.3.3 herein), shall be prepared in accordance with the requirements of MIL-C-38778 for Class I, type 6 checklists. Separate checklists shall be prepared for each control-monitor. Mission data cards (see 3.2.5.2.2 herein) shall be included in the checklists, if applicable. Only the most critical notes, cautions, and warnings shall be included in the checklists.

3.2.5 Content (Volume III). Volume III (Mission Planning) shall contain planning procedures and sample problems for each applicable delivery mode, using a separate section for each mode. Within each section, the data shall include mission description, planning procedures and sample problems. Ballistic tables for applicable bombs will be published in a separate manual supplement to Volume III. Title of the supplement shall be "Ballistic Tables" and identification of bombs covered.

3.2.5.1 Tables and Charts. Bomb table pages in the supplement manual shall be identified as specified in 3.1.1.4 herein.

3.2.5.2 Arrangement (Volume III). This volume shall have the following arrangement:

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Section I - General Mission Planning Information

Section II, III, etc. - Mode Mission Planning

Part 1 - Mission Description

Part 2 - Planning Procedures and Sample Problems

Part 3 - Planning Forms

3.2.5.2.1 Section I - General Mission Planning Information. This section shall describe the use of charts that are common to all delivery modes; sample charts only shall be used in this section.

3.2.5.2.2 Section II, III, etc. - (Mode) Mission Planning. One section shall be used for each delivery mode. Each mission planning section shall contain the following information:

Part 1 - Mission Description. This part of the section shall describe the mode of delivery, bombs applicable to this mode and the parameters of data presented.

Part 2 - Planning Procedures and Sample Problems. Reasonable mission parameters shall be assigned for a problem. The planning procedures shall be demonstrated for one bomb only.

Part 3 - Planning Forms. This part of the section shall contain all planning forms, mission data cards, and necessary charts.

3.3 Single-Volume Nuclear Weapon Delivery Manuals. All data may be presented in a single volume, rather than the multivolume arrangement, if authorized. Single-volume manuals are normally used for delivery TOs for non-US use.

Front Matter

Section I - Description

Section II - Normal Aircrew Procedures

Section III - Emergency Aircrew Procedures

Section IV - Planning Procedures and Sample Problems

Section V - Planning Charts and Tables

Section VI - Supplementary Data

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Alphabetical Index

If no data are available for a particular section, that section shall be omitted and succeeding section renumbered. Ballistic tables may be published in a separate manual supplement to this manual.

3.3.1 Content. Content of each section and the front matter shall be similar to that used for and following the requirements of applicable portions of the multivolume manual. In sections II and III, the delivery mode and/or bomb designations and other pertinent information shall be placed below the TO number. Bomb tables shall be tabbed and pages identified as specified in 3.1.1.4 herein. Section VI shall contain bomb effects and escape data, plus any other required supplemental information. Specific procedural headings may be omitted, added, or reworded to fit operational requirements. For examples, Weapon Acceptance Procedures, shall be used instead of the heading "Preflight Procedures."

3.4 Practice Bomb Technical Manual. Technical manuals for the delivery of practice bombs shall normally be single-volume manuals with checklist(s). They are titled Aircrew Practice Bomb Delivery Manuals; titles may include the delivery modes.

3.4.1 Scope. This technical manual shall contain the information necessary to plan and practice a simulated nuclear bomb delivery with nonnuclear bombs. This information shall include armament controls, bomb descriptions, procedures for the use of practice bombs, and error analysis. Ballistic tables (or charts) will be published in a separate manual supplement to this manual. Title of the supplement manual shall be "Ballistic Tables" and identification of bombs covered.

3.4.2 Content and Arrangement. The practice bomb delivery manual shall contain information and procedures to plan and execute a simulated nuclear bomb delivery, with practice bombs or dispensers. Armament control and release system information, bomb descriptions, ballistics, mission planning and delivery procedures shall be included. The manual shall normally include the following:

Front Matter

Section I - Description

Section II - Normal Aircrew Procedures

Section III - Planning Procedures and Sample Problems

Section IV - Planning Charts and Tables

Section V - Error Analysis (can be included with each mode)

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Alphabetical Index

If no data are available for a particular section, that section shall be omitted and succeeding sections renumbered. Normal aircrew procedures shall parallel those used for the parent (nuclear) bomb, although the procedures may be broken out by bomb rather than by aircraft monitor and control. Terminology shall follow that used in the applicable nuclear bomb delivery manual. Mission planning may be presented as a separate section for each delivery mode; each section would parallel the content of applicable sections of Volume III of the parent weapon (nuclear) manual. In this case, Sections II and III would be replaced by one section for each delivery mode covered, using as many sections as required and renumbering succeeding sections. Error analysis for each delivery mode shall be included in the last section (or may be included at the end of each delivery mode). Separate checklists are normally required for dispenser-released bomblets and for practice shapes.

3.5 Maintenance of Manual(s) and Checklist(s). The manuals and checklists shall be maintained in accordance with the requirements of MIL-M-38784. Change and revision requirements shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Manuals and checklists. (Basic and Revisions)

4.1.1 In-Process Review. Draft manuscripts shall be reviewed as scheduled during the writing process. Comments and corrections resulting from this review shall be incorporated into the review copy or preliminary TO. The provisions of TO 00-5-1 for validation shall apply.

4.1.2 Review Conference Verification. Unless otherwise specified in the contract, a review conference shall be held at the contractors facility, or if combined with a verification, at an operational location. Review copy shall normally be in the form of a preliminary TO, with text double spaced and containing all required art. The manual shall be reviewed/verified for the technical adequacy and accuracy. Classification markings, if required, shall also be reviewed at this time.. Before preparation of reproducible copy is begun, a final manuscript review may be required to insure incorporation of all review/verification inputs.

4.1.3 Prepublication Review. Reproducible copy of the manual and checklist (if required) shall be reviewed by the Government as contractually specified for technical adequacy and accuracy and for conformance to specifications. Reproducible copy shall meet the requirements of MIL-M-38784.

4.2 Changes. Unless otherwise specified in the contract, all changes shall receive in-process and prepublication reviews. Extensive changes (such as addition of a new weapon capability) may require verification. If so, the provisions of 4.1.2 here shall apply.

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4.3 Negatives. Negatives shall meet the requirements of MIL-P-38790.

4.4 Reproduction Assembly Sheets. Reproduction assembly sheets shall be prepared as specified in MIL-P-38790.

5. PREPARATION FOR DELIVERY

5.1 Negatives shall be prepared for shipment as required in MIL-P-38790. The provisions of MIL-P-38790 relative to use of reproduction assembly sheets with negative shipments shall apply.

6. NOTES

6.1 Intended Use. Technical manuals and checklists prepared according to this specification are for use of aircrews in the delivery of nuclear bombs and practice bombs for tactical fighter aircraft.

6.2 Ordering Data. Procurement documents should specify:

a. Number, title, and date of this specification.

b. Data required (no. of volumes, TO nos., aircraft/weapon designations, checklist requirements).

6.3 Manual Designations. Although TO numbers are furnished through the procuring activity, nuclear bomb delivery TO numbering practices are provided for information. TOs are numbered as 1F-XXX-25-X, with dash 25-1 for Volume I, dash 25-2 for Volume II, checklists as dash 25-2CL-X, and dash 25-10, with checklists as dash 25-( )CL-( ). Single volume manuals for non-US use are numbered as dash 25-4, with dash 25-4CL-X checklists. Practice bomb delivery manuals for non-US use are numbered as dash 25-4-10, with dash 25-4-10CL-X checklists. Supplement manuals containing ballistics tables will be numbered dash 25-3-1, dash 25-10-1, or dash 25-4-10-1, as applicable.

Custodian
Air Force - 15

Preparing Activity
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Review Activity
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-27579E	Manuals, Technical: Aircrew, Nuclear Weapon (BOMB) Delivery, M

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)* mode
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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